AI-Supported Adaptive Training Programs in Professional Football

Personalized Training Programs Based on Data through Artificial Intelligence

MSc Ebru Kemikkiran

02.02.2025
Computer Science with Software Engineering
Wrexham University

ABSTRACT

This study explores the integration of artificial intelligence (AI) and machine learning technologies into professional football training systems to develop adaptive, personalized programs. Using data collected from wearable sensors such as GPS, heart rate monitors, and accelerometers, AI systems can monitor players' physical performance in real time and generate individualized training recommendations. The research reviews current academic studies, machine learning algorithms, commercial systems, and the technical architecture necessary for real-time data processing and feedback. Additionally, it emphasizes the importance of tailoring training based on positional differences, such as the varying needs of goalkeepers, defenders, midfielders, and forwards. The proposed system aims to optimize load management, prevent injuries, and enhance player performance by combining long-term training plans with short-term dynamic adjustments.

Keywords: *Artificial Intelligence, Machine Learning, Adaptive Training, Football Analytics, Wearable Technology, Injury Prediction, Position-Specific Training, Real-Time Feedback, Performance Optimization, Sports Technology*

INTRODUCTION

- In modern football training, artificial intelligence (AI) and machine learning are employed to create individualized training programs based on each player's unique performance data. Large volumes of data such as location, speed, heart rate, and acceleration—collected through wearable GPS sensors, heart rate monitors, and other devices—are analyzed by AI to design training plans tailored to the current needs of each athlete.

I. PERSONALISED, DATA-DRIVEN TRAINING PROGRAMS USING ARTIFICIAL INTELLIGENCE

In modern football training, artificial intelligence (AI) and machine learning are utilized to create personalized programs based on each player's individual performance data. Large volumes of data, such as location, speed, heart rate, and acceleration collected via wearable GPS sensors, heart rate monitors, and other devices, are analyzed by AI to design training plans tailored to the athlete's current needs [1]. These systems assess a player's past performance trends, physical characteristics, and learning styles to develop targeted and customized training loads. For example, AI can identify areas in which a specific player needs improvement and adjust the content of the training accordingly [2].

With personalized AI-supported programs, workload and recovery times can be optimized, resulting in dynamic training regimens that adapt in real time to the athlete's progress. This approach aims to maximize each player's potential while minimizing the risks of overtraining or injury. As noted by the company Playermaker, AI-generated, highly personalized programs are structured to match the exact needs of the athlete, leading to improved performance and preventing issues caused by excessive loading. AI's ability to detect harmful training patterns early enables coaches to proactively adjust training plans [3]. As a result, rather than relying on traditional, one-size-fits-all models, it becomes possible to implement adaptive, data-driven training prescriptions tailored to each individual player [2].

II. FUNCTIONING REAL-TIME AND PERIODICALLY UPDATED ADAPTIVE TRAINING SYSTEMS

Adaptive training systems operate through a closed-loop feedback mechanism: the data collected from the player is continuously monitored, and the training program is updated based on this feedback. For example, metrics such as heart rate, running distance, and number of

sprints can be tracked in real time during training to assess the athlete's current condition. AI algorithms can detect changes in the player's performance on that particular day and instantly adjust the training intensity accordingly [4]. In a fitness app example, it has been observed that AI continuously monitors the user's performance and automatically adjusts the intensity of exercises: as the athlete becomes stronger, the difficulty increases; when signs of fatigue are detected, the intensity is reduced to ensure a safe and effective workout [3]. Similarly, in team sports, AI provides coaches with real-time alerts and suggestions, allowing for immediate or post-session adjustments to the training.

At the core of real-time adaptation lies the instant processing of collected data and its comparison against predefined targets. These systems typically monitor players' momentary physical conditions to detect signs of fatigue or overload early and adjust the training dose accordingly [3]. For instance, if a player's heart rate remains above the expected range and recovery time increases, the adaptive system may decide to reduce the intensity of the next day's session. Wearable systems like STATSports integrate real-time movement data and other biometric indicators to provide early warnings to coaches, enabling quick adjustments in training intensity or duration to keep the athlete within the desired performance range [3].

In a study on an AI-powered feedback controller, it was demonstrated that the system could automatically optimize future training loads to minimize the discrepancy between planned goals and the athlete's actual performance. Such an intelligent controller significantly reduced deviations from the training plan compared to random or fixed-ratio control methods, helping maintain the targeted load profile. Ultimately, real-time adaptive systems aim to continuously monitor data and make minute-by-minute adjustments to deliver the most appropriate training load based on the athlete's current condition [5]

III. MACHINE LEARNING ALGORITHMS AND APPLICATIONS THAT CAN BE USED

Various machine learning algorithms can be applied in AI-assisted training programs for different purposes. The choice of the appropriate algorithm depends on the specific problem to be solved, and some common examples include:

- Decision Trees and Random Forests: These algorithms are particularly useful for classification and prediction tasks. In sports science, decision tree-based models have been successfully used, for example, to predict the injury risk of players. A review study noted that models utilizing decision tree algorithms could predict injury risk with over 66% accuracy [6]. Decision trees are also valuable in forecasting training strategies; as emphasized by the company Catapult, decision tree-based approaches are widely used in sports analytics for strategy prediction [7]. Since these models can present key

thresholds and conditions in a way that is understandable to humans, they can also provide rule-based recommendations for training loads or rest decisions.

- Artificial Neural Networks (ANNs): Neural networks are effective in identifying patterns from complex and high-dimensional data. In particular, recurrent neural networks (RNNs, including LSTM models) are suitable for analyzing time-series data (e.g., daily performance and recovery data) and can predict a player's future performance or fatigue level [8]. In fact, a trained RNN model has been shown to predict a player's physical readiness before a training session and alert the coach accordingly, thus helping to optimize the session. Convolutional neural networks (CNNs), on the other hand, are prominent in analyzing visual or spatial data and are frequently used in sports for video analysis or recognizing patterns in sensor signals. For instance, a study in tennis demonstrated that body language during matches could be analyzed using CNNs to determine players' emotional states. Coaches were then able to respond in real time to stress or frustration by adjusting training loads or tactics accordingly. This image-based CNN application is an innovative example of how emotions can be interpreted and incorporated into training strategies [1]. Overall, deep learning techniques (ANN, RNN, CNN) provide advanced insights into performance prediction, player classification, and technical-tactical analysis by learning complex, multidimensional relationships.

- Reinforcement Learning (RL): These algorithms treat training planning as a decision-making process and learn the best strategies through trial and error. In reinforcement learning, an "agent" decides which type of training to prescribe each day, receives feedback in the form of rewards or penalties, and gradually learns which training sequence leads to optimal performance improvements or desired outcomes. Theoretically, RL is a highly suitable tool for optimizing sequential decision-making in training plans. One study showed that a personalized fitness program developed using RL could dynamically adjust the intensity, duration, and frequency of exercises based on the athlete's performance and feedback [9]. The system thus learns on its own how to optimize each training session in line with long-term goals. For example, an RL model could define "winning the weekend match" as a reward and learn to adjust the training loads during the week to maximize that outcome.

- Clustering and Recommendation Systems: Unsupervised learning techniques can be used to group players or conditions with similar characteristics. For example, clustering algorithms can be applied to players' data such as running distance, number of sprints, or heart rate patterns to identify subgroups with similar load profiles. This allows for the development of tailored training protocols for each group (e.g., high sprinters vs.

endurance-focused players). Machine learning-based recommendation engines can use historical data and similar player profiles to suggest the most appropriate training content for a given athlete [10]. For instance, a running app may analyze users' performance data and goals to generate personalized training recommendations inspired by programs that have led similar athletes to success. These systems often employ collaborative filtering or content-based filtering techniques.

In addition to the algorithms mentioned above, methods such as support vector machines (SVMs), genetic algorithms, and Bayesian models also find application in sports data analysis. What matters most is selecting the algorithm that best fits the problem and developing a reliable model accordingly. For example, Catapult identifies decision trees as commonly used for strategy forecasting, neural networks for performance analysis, and clustering techniques for player segmentation in the context of sports analytics [11]. In conclusion, the rich array of machine learning algorithms enables the development of specialized solutions for various applications ranging from performance prediction and injury risk assessment to training optimization and data-driven decision support.

IV. EXISTING ACADEMIC STUDIES, APPLICATIONS, AND MARKET EXAMPLES

There are numerous successful applications of AI-based adaptive training systems both in academia and in the sports industry. On the academic side, recent research has focused on topics such as injury prediction and training load optimization. For example, the METIC model developed in 2023 uses deep learning to analyze players' historical match and injury data and predict future injury risks. Leveraging the transformer architecture originally developed in natural language processing, the model identifies patterns in sequential data, such as the likelihood of an Achilles tendon rupture, and alerts coaches and medical staff in advance. This enables preventive measures for high-risk players, such as increased rest, load reduction, or strengthening exercises [1].

Similarly, in the field of machine learning-based injury prediction, the startup Zone7 uses in-game and training data to provide teams with proactive alerts regarding player injury risk [1]. By processing GPS data, acceleration, and signs of fatigue, Zone7 can, for example, issue predictions such as "Player A is at high risk for a muscle injury next week if overloaded." Clubs then use these insights to reduce the training load or implement extra recovery protocols. These academic and commercial efforts form the scientific foundation of adaptive training systems and help validate their effectiveness [1].

In the marketplace, several systems are widely used by professional teams:

- STATSports: One of the world's leading GPS-based player tracking systems, STATSports collects and analyzes players' training and match data to monitor performance. Its wearable devices (e.g., the Apex unit) record real-time data on speed, distance, acceleration, and heart rate. These data are uploaded to software for detailed reporting after each session. Recently, STATSports has integrated advanced AI algorithms into its devices. The new Apex model, introduced in 2023, includes a more powerful processor capable of real-time motion pattern analysis and provides around 70 new real-time metrics compared to previous generations. According to the company's CEO, this upgrade allows for automatic on-device AI analysis, enabling coaches to focus more on players rather than processing raw data [12]. For instance, if the system detects unusual fatigue or a performance drop during training, it can immediately suggest an intervention. The software also generates weekly and monthly load reports, providing a scientific basis for training planning. In one case, STATSports data was used throughout a season to track players' acute-to-chronic workload ratio (A:C), and training was gradually tapered before matches to keep the ratio within the optimal range of 0.8–1.5 [13]. This confirmed the effectiveness of data-driven periodization and demonstrated how training plans could be adjusted and optimized in real time based on objective metrics [13].

- Catapult Sports: Another industry leader, Catapult offers wearable tracking devices and integrated analytics software. Its GPS vests, such as the Vector system, accurately measure key metrics like distance covered, speed, acceleration, and workload. Coaches can monitor each player's load individually and tailor training plans as needed [14]. Devices like the S7/T7 series provide detailed data to support personalized training and injury prevention strategies. Catapult also integrates video analysis with performance data, offering deep insights [14]. For example, syncing match footage with GPS data allows for simultaneous evaluation of a player's tactical position and physical workload. According to Catapult, the fusion of video and data supports a fully data-driven approach to player development and is essential in modern training systems [14].

- Metrica Sports: Specializing in video analysis, Metrica Sports provides teams with automated player tracking data and advanced video processing tools. Its platform uses AI to analyze training and match videos, extracting tactical data such as player positioning, distances between players, and team shape integrity. The combination of detailed visual analysis and data analytics allows coaches to precisely evaluate player movements and adapt tactics or training sessions accordingly [15]. For instance, if analysis reveals that defenders are leaving too much space between them, this insight can inform targeted drills in subsequent training. Metrica's intuitive interface and

visualization tools make it accessible not only to elite clubs but also to amateur and semi-professional teams seeking to incorporate data and video analytics into training planning [15].

- Other Systems and Applications: In addition to the examples above, many innovative wearable devices contribute to the adaptive training ecosystem. The WHOOP wristband, for example, tracks internal load parameters such as heart rate, heart rate variability (HRV), and sleep quality throughout the day [1]. It analyzes this data using cloud-based AI to provide a recovery score and personalized daily training recommendations. Athletes receive color-coded feedback—such as “green” (ready) or “red” (fatigued)—and can adjust their training accordingly [1]. Similarly, Intel’s 3D Athlete Tracking (3DAT) system uses cameras and AI to analyze movement in three dimensions in real time. It captures data on running technique, stride length, and joint angles to help improve technical form. In football specifically, devices like Playermaker, which attach to the ankle, collect detailed technical data such as number of ball touches, use of both feet, and step symmetry. Using AI, the device then recommends drills to improve technique [15]. In the endurance sports domain, apps like TrainAsONE and Xert offer continuously adaptive training plans for amateur runners and cyclists, demonstrating how the concept of AI-driven training is becoming widespread across all levels [15].

In summary, both academic research and commercial products have established the practical viability of AI-assisted training. Teams are able to monitor training data in real time with STATSports and Catapult, integrate tactical and technical insights using Metrica, and take proactive steps to protect player health with tools like Zone7. This multifaceted ecosystem has elevated professional training planning to a more scientific, data-driven, and personalized level.

V. INTEGRATING WEEKLY TRAINING PLANS WITH REAL-TIME FEEDBACK

In football, a microcycle (typically a week-long training plan) involves pre-planning the distribution of physical loads and rest days throughout the week. However, real-time data collected from players may necessitate fine-tuning of this plan. The key to integration lies in establishing a balanced interaction between long-term planning and short-term feedback. In other words, while weekly goals and periodization provide a framework, the player’s day-to-day condition and training responses allow for adjustments within that framework [13].

In practice, integration works as follows: at the start of the week, coaches prepare a weekly training load schedule based on the match calendar (e.g., strength training on Monday, high-

intensity training on Tuesday, active recovery on Wednesday, etc.). This plan follows scientific periodization principles and aims to have players in peak condition by the weekend [13]. For example, as match day approaches, training intensity typically tapers down, while midweek sessions may include higher loads to maintain fitness. As the week progresses, GPS and biometric data collected from players continuously monitor how well the plan is being followed. Suppose Wednesday's data indicates that a key player is more fatigued than expected, has had poor sleep, or hasn't fully recovered from the previous session—an adaptive system could use this information to lighten Thursday's training session. This prevents overexertion and better prepares the player for the weekend match.

Adaptive algorithms aim to align such micro-level changes with the broader macro plan. For instance, in a study by Connor et al., an AI-driven feedback system monitored daily training loads within a 12-week program and automatically adjusted future session intensities if there were deviations from target goals [5]. Even when unexpected events occurred (such as injury, illness, or early termination of training), the system was able to re-optimize the remaining sessions to stay on course with the overall training objectives. Results showed that this approach significantly reduced deviations from the planned workload compared to traditional methods [5]. Essentially, the system adheres to the goal of achieving the "total weekly training load" while making necessary daily adjustments.

Another important approach is integrating scientific metrics such as the Acute:Chronic Workload Ratio (A:C). Weekly planning must ensure that a player's acute (1-week) training load remains within a safe range relative to their chronic (4-week) load average. Feedback systems calculate this ratio in real time to verify whether the training is progressing as planned. If a player's A:C ratio exceeds the upper threshold of 1.5 (indicating the past week's load is much higher than usual), the system may suggest reducing the workload for the rest of the week [13]. Conversely, if the load is below target, it may recommend additional individual sessions. This ensures that periodization principles are upheld with adjustments informed by real-time data.

Today, some AI-driven platforms automate this integration process. For example, Xert, an app used in running and cycling, offers a continuously adaptive training schedule instead of a fixed program. Its Adaptive Training Advisor evaluates the athlete's condition after each session and updates the next day's session accordingly. As a result, scenarios like "I couldn't follow today's plan" or "I feel great today, what should I do?" are predicted and accommodated within the system. Such platforms make feedback integration user-friendly and accessible, even for amateur athletes. At the professional level, this integration is usually overseen by performance scientists and coaches. AI systems generate recommendations, and humans make the final decision [10]. For example, if an AI system reports, "Player X completed 20% less high-speed running today than planned; consider adding sprint drills tomorrow," the coach then considers

the match schedule and the player's subjective condition before acting on the suggestion.

In summary, combining a weekly plan with real-time data feedback has become a modern necessity in sports science. Static plans are becoming obsolete, giving way to dynamic, predictive strategies. While the weekly program provides a roadmap, real-time feedback systems continuously fine-tune that path, keeping the athlete on the safest and most efficient trajectory [10]. This approach ensures that both the big picture (season goals, match performance) and the daily details (fatigue, motivation) are managed simultaneously.

VI. PROPOSED TECHNICAL ARCHITECTURE AND COMPONENTS

To implement such an adaptive training system, a multi-layered technical architecture must be designed. In general, this architecture consists of the following components: data collection layer, analysis and processing layer, recommendation engine, and user interface:

- Data Collection Layer:

  This bottom layer gathers data from all sensors and devices worn by the player. These may include GPS vests, accelerometers, gyroscopes, heart rate monitors, sleep trackers, and possibly ball or ankle-mounted sensors used during training. The raw data from these devices (e.g., position coordinates per second, heart rate counts, step counts) is transmitted wirelessly to cloud servers either in real time or at the end of the training session. For example, the STATSports Apex device can stream data via Bluetooth or ANT+ to a receiver on the sidelines, while a wearable like WHOOP syncs data continuously to the cloud throughout the day [12]. What matters most at this layer is the synchronized and compatible collection of data from different sources and its secure storage [12]. All data is stored in a data warehouse or a time-series database.

- Analysis & Data Processing Module:

  At the middle layer, raw data is transformed into meaningful insights. The first step involves cleaning and synchronizing the data (e.g., aligning GPS and heart rate data by timestamp, filtering missing or erroneous entries). Then, business intelligence and data analytics processes are applied to extract derived features from the raw data. Key performance metrics such as total running distance, high-speed distance, number of sprints, average heart rate, heart rate variability during training, and RPE (Rate of Perceived Exertion) are calculated. In the STATSports system, such metrics, including total distance, high-speed running (HSR), and dynamic stress load (DSL) [13], are automatically generated and reported. Within this module, the core AI algorithms

operate. Machine learning models process the incoming data to make predictions and assessments:

  - An injury risk model can generate a risk score based on recent training loads and recovery trends.
  - A performance prediction model can estimate a player's readiness for match day based on current training data.
  - An anomaly detection system can flag unusual fatigue or performance drops.

- These analyses may run in real time or after each training session. For instance, the system may assess whether a player has progressed or regressed compared to the previous day, using machine learning algorithms to benchmark the results. This module supports the decision-making process by generating insights using the machine learning techniques outlined in Section 3. In complex, multi-source architectures, AI-based interpretation—as highlighted in the TTMS study—can reveal intricate patterns that human analysts might miss. The analysis module is also responsible for comparing calculated metrics and model outputs to predefined thresholds (e.g., calculating the Acute:Chronic Workload Ratio) [1].

- Recommendation Engine (Decision Support System):

  The insights from the analysis module are translated into actionable training suggestions. This layer houses rule-based logic and optimization algorithms. For example, if the injury risk model shows an 80% risk for a player, the recommendation engine might suggest "Assign rest day tomorrow." Or, if a player's high-intensity running distance over the past three days is below the plan, the system may recommend "Add extra interval training."

  The recommendation engine also takes the coach's predefined goals and constraints into account. For instance, if Friday is intended to be a light session, the system will not suggest adding extra load that day. This engine operates using a blend of business rules (if-then logic) and trained models. In advanced systems, this module may even be driven by a reinforcement learning agent that learns over time which recommendations yield the best results. However, in practical settings, the system typically provides a shortlist of recommendations.

- User Interface (UI) and Feedback:

At the top of the architecture is the interface through which coaches, performance staff, and possibly players interact with the system. This interface may be a web-based dashboard, a mobile app, or a specialized tablet application. It visually presents information from lower layers through charts, tables, alerts, and recommendation lists. For instance, the UI might highlight in red that a player has reached 90% of their running distance goal but only 50% of their high-speed running target, with a note such as “high-speed workload low.” This allows coaches to instantly grasp the situation. Through the interface, the coach can approve or modify system suggestions. For example, if the system suggests a rest day but the coach wants to schedule a light technical session instead, this can be manually adjusted—and the system will learn from it.

A well-designed UI consolidates the weekly schedule, daily adjustments, and player status reports into one view, simplifying the decision-making process. A separate sub-interface may also be provided for players to track their own metrics and receive personalized feedback (e.g., “You exceeded your running goal today” or “Your heart rate was within normal limits”).

In implementing this architecture, several technical considerations are essential:

- Real-time capability: The system must process data with minimal latency.
- Scalability: It should handle data streams from the entire team without performance issues.
- Reliability and security: Data loss must be prevented, and sensitive health information must be protected.
- Integration: The system must be compatible with different devices and data formats.

Most modern systems are cloud-based, meaning data is processed and stored on internet servers. However, as seen in newer STATSports devices, on-device AI (edge computing) is becoming a trend. This allows real-time analysis directly on the device—particularly useful for instant feedback during training—without needing to transmit data to the cloud first [12]. Still, a centralized database and analysis platform remain indispensable for long-term data management and complex modeling.

Lastly, for this architecture to succeed, it must operate in continuous collaboration with the

technical staff. Recommendations and alerts generated by the system must be filtered through human expertise. While the architecture offers automation, the role of the human coach is vital—to interpret insights, define goals, and make final decisions. AI assists, but human judgment remains central.

VII. CONSIDERING THE SPECIFIC NEEDS OF DIFFERENT FOOTBALL POSITIONS

When designing adaptive training systems, it is essential to recognize that players in different football positions have distinct physical and technical demands. A goalkeeper's workload profile differs greatly from that of a midfielder, and their training focuses and risks also vary. An AI-assisted system can account for these positional differences in several ways:

- Goalkeepers:

  Goalkeepers typically cover much less distance during matches but perform more explosive actions (e.g., quick dives, jumps for crosses). According to one study, a goalkeeper runs about 5.5 km on a match day, performs only 50–100 meters of sprinting, and executes 5–10 high-intensity accelerations or decelerations. These values are significantly lower than those of a midfielder. Therefore, an adaptive system should apply different workload thresholds for goalkeepers. Instead of tracking running distance, it could monitor reaction times, jump height, and kicking range. If a goalkeeper completes a high number of explosive actions (e.g., 15+ dives) in a session, the system might suggest rest or low-intensity technical training the next day. Due to the nature of their role, goalkeepers also require upper body strength and flexibility [16]. The adaptive program should periodically emphasize exercises that develop these attributes, such as plyometrics and mobility training. Since goalkeepers may remain stationary for long periods and then need to react explosively, mental focus drills and active recovery techniques may also be integrated.

- Defenders:

  Defenders can be further divided into center-backs and full-backs. Center-backs perform short, explosive movements (e.g., tracking strikers, sudden changes in direction), while full-backs tend to cover more distance with up-and-down runs along the flanks. Overall, defenders cover a moderate-to-high total distance (~9–11 km per match, with full-backs on the higher end). Adaptive systems should closely monitor acceleration and sudden speed changes for defenders, especially center-backs, since timing and quickness are critical. If a center-back's number of high-intensity accelerations falls below target, the

system may recommend additional speed drills. For full-backs, endurance and speed are key, so the system should track their high-speed running distance. Strength training is also vital for defenders due to frequent physical duels; adaptive plans may incorporate more resistance training for defenders compared to other positions. For instance, two weightlifting sessions per week might be standard for a defender, while only one may suffice for an attacker. An AI-based system can analyze a defender's in-game stats (e.g., aerial duels won, tackle success) and, if performance drops in those areas, increase the strength training volume accordingly.

- Midfielders:

  Midfielders are typically the players who cover the most distance and have the highest aerobic and repeated sprint demands. Elite-level midfielders may run 11–13 km per match, much of it at moderate intensity, interspersed with sprints. Adaptive systems must pay special attention to recovery capacity for midfielders, as they are often the most fatigued during congested match schedules. For example, if a midfielder's morning heart rate variability (HRV) is lower than usual (a sign of fatigue), the system may reduce training load for the day. Midfielders also require a broad range of skills, since they contribute to both defensive and offensive phases. An adaptive program should aim to develop speed, endurance, and agility in a balanced manner. While analyzing midfielder data, the system might not prioritize maximum speed as much as for attackers but rather focus on total load and intensity distribution. Nevertheless, if a midfielder's high-speed running distance is consistently low, the system may interpret it as a lack of explosive speed and recommend targeted sprint training. Since passing accuracy is also crucial for midfielders, the system could optimize technical sessions accordingly—for instance, planning additional passing drills if a midfielder's pass completion rate drops in matches.

- Forwards and Wingers:

  Attacking players (forwards and offensive wingers) may run slightly less than midfielders overall, but they perform the highest number of high-speed efforts. A forward may cover 9–11 km per match, with key actions involving explosive sprints for scoring chances or pressing. Adaptive systems should track metrics such as number of sprints, maximum speed, and explosive power. If a winger's top speed during training is below their known potential, the system may recommend more speed work. For forwards, shooting frequency and accuracy are also important indicators; some systems can even track shot efficiency (e.g., percentage of accurate shots). If a forward hasn't had enough

scoring opportunities in recent games and sprint metrics are down, the system may interpret this as a physical issue (e.g., lack of freshness or sharpness) and reduce training load to allow for recovery. Agility and quick changes of direction are also vital for these players; systems can monitor 5–10 meter acceleration data or zigzag run tests to assess form. Position-specific settings may lead the adaptive algorithm to assign more speed and agility drills to attackers, while defenders may receive relatively less long-distance running.

Adaptive systems can also incorporate positional differences during the modeling phase. Machine learning models can treat position as an input feature. For example, an injury prediction model might use different thresholds for goalkeepers than for midfielders. Likewise, rule-based thresholds can be tailored to position—for instance, a rule like "If a goalkeeper exceeds 20 km of running in a week, issue a warning" would be irrelevant for midfielders but critical for goalkeepers. Research has shown that the relationship between external training load and physical characteristics of players depends heavily on their tactical role (i.e., position). This underscores the importance of position-specific approaches in adaptive systems [17]. For example, a tall, powerful center-back and a short, fast winger differ not only in their match roles but also in their training requirements; a well-designed system can recognize and accommodate these differences.

In conclusion, adaptive training systems must function with distinct parameters and objectives tailored to goalkeepers, defenders, midfielders, and forwards. The universal aim across all positions is to maximize performance and minimize injury risk, but the path to achieving this must be position-specific. Sharpening a goalkeeper's reflexes, increasing a defender's endurance and strength, maintaining a midfielder's aerobic capacity, and enhancing a forward's explosive speed and finishing—these can all be achieved within the same system via separate modules and configurations. A well-structured AI-powered training platform will treat positional differences as a personalization layer, ensuring each player receives the most suitable workload according to their role. In turn, this enables every part of the team to perform at its best—raising the overall level of team performance to its highest potential.

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